Course Notes

Lecture 14 — Course Review and Mini-Project Demonstration

Communication Technologies for the Internet of Things

Academic year 2026–2027

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Text to support independent study

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How to Use These Notes

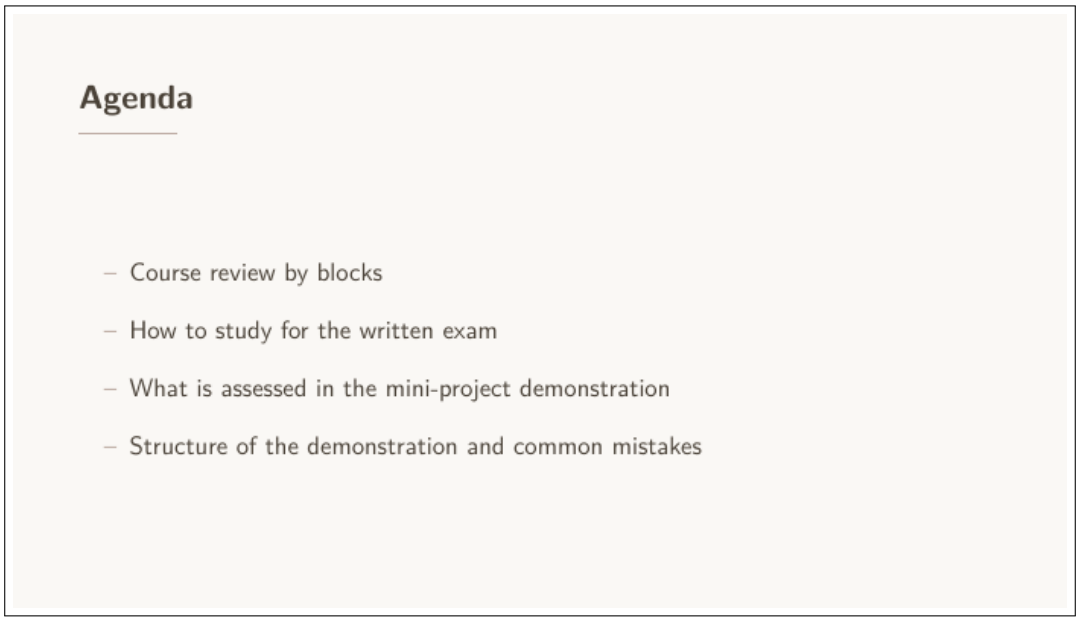
These notes follow the lecture slide by slide. Each page reproduces a slide and states, below it, what should be taken from it.

Reading them does not replace the lecture or the slides. They are meant for later study, once the student has heard the lecture and wants to consolidate, check that the essentials were understood or prepare for assessment. The comments make explicit links between slides that remain implicit in the oral presentation, and point out where the material is commonly misunderstood.

Whenever a comment poses a question, that question is the verification criterion: if you cannot answer it, go back to the slide before moving on.

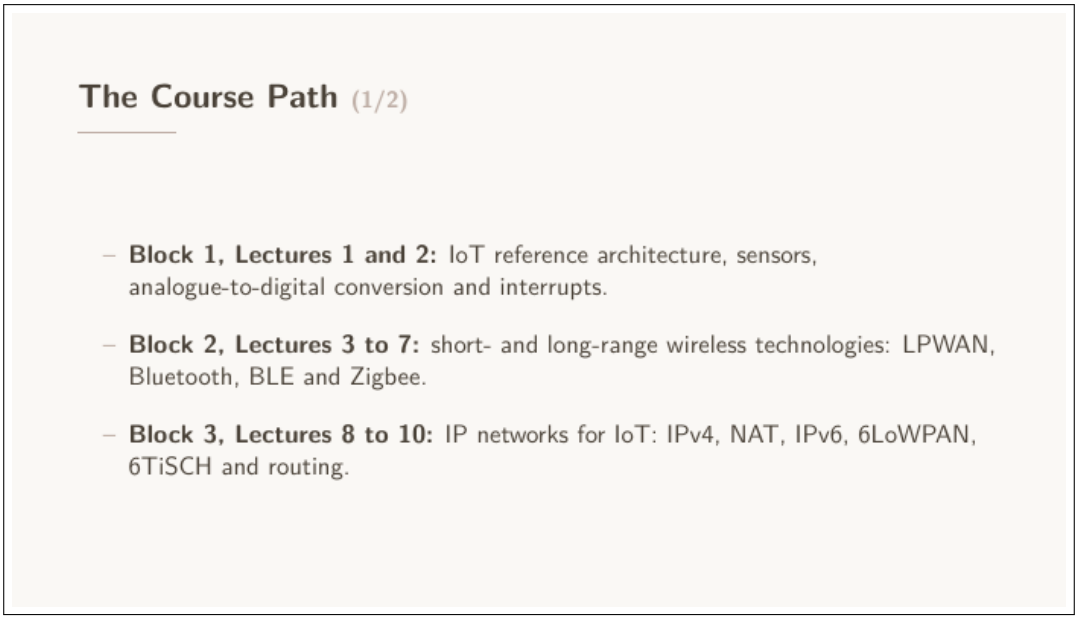
1. Commentary on the Slides

Slide 2. Agenda



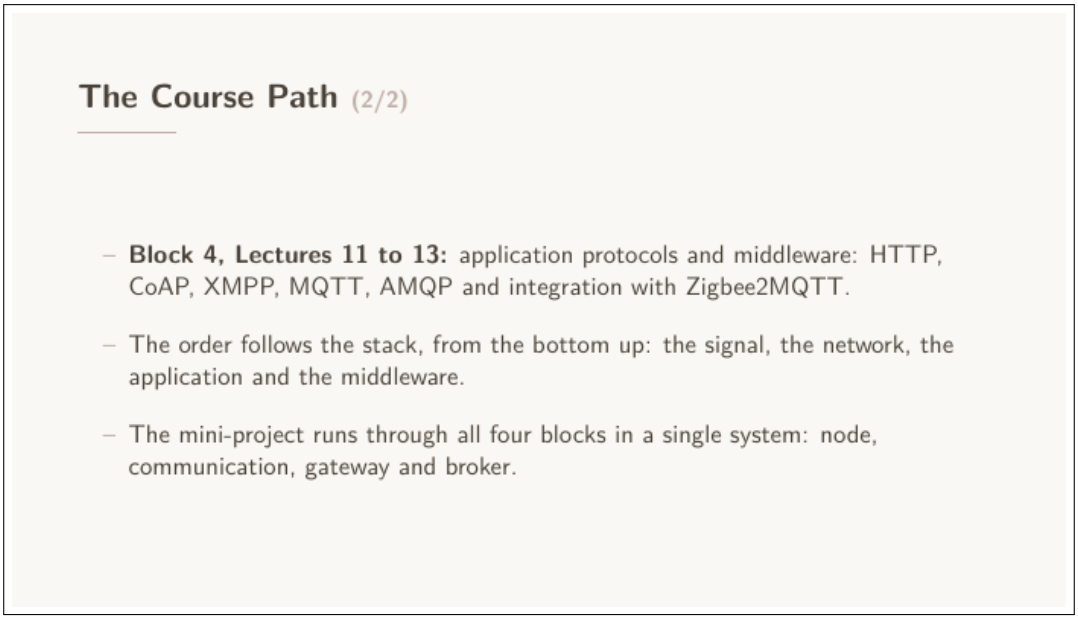
The lecture. The lecture has two functions. The first is to consolidate the material, block by block, and prepare for the written exam. The second is the mini-project demonstration, in which each group shows the system working and answers questions about it.

Slide 3. The Course Path (1/2)



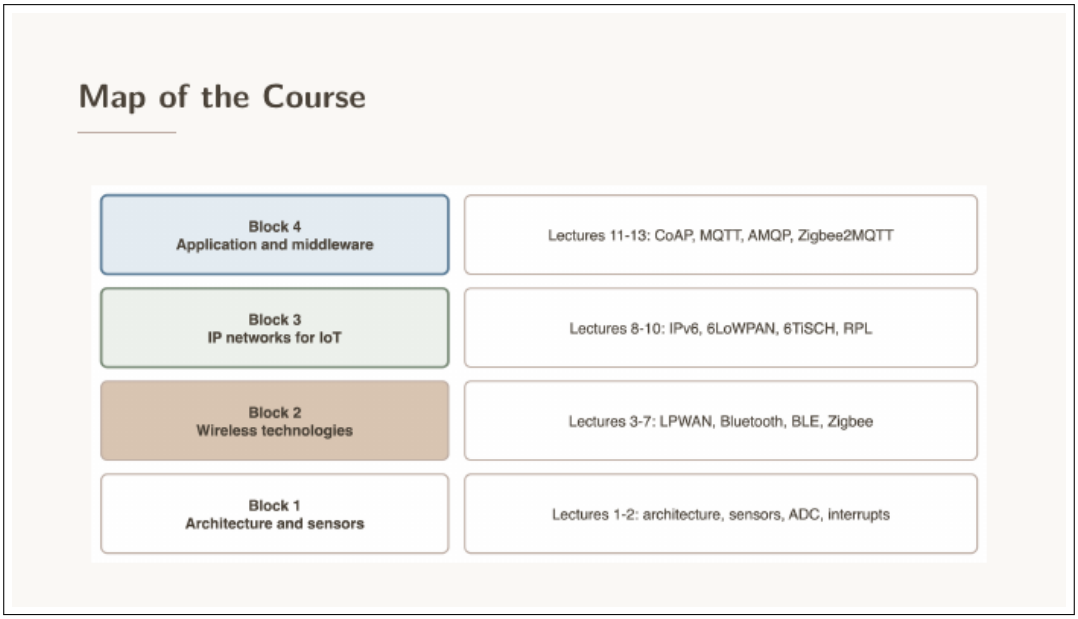
The first two blocks. The first block deals with architecture and sensors. The second deals with wireless technologies, from short to long range. Both sit at the base: without a reliable signal, nothing that follows works.

Slide 4. The Course Path (2/2)



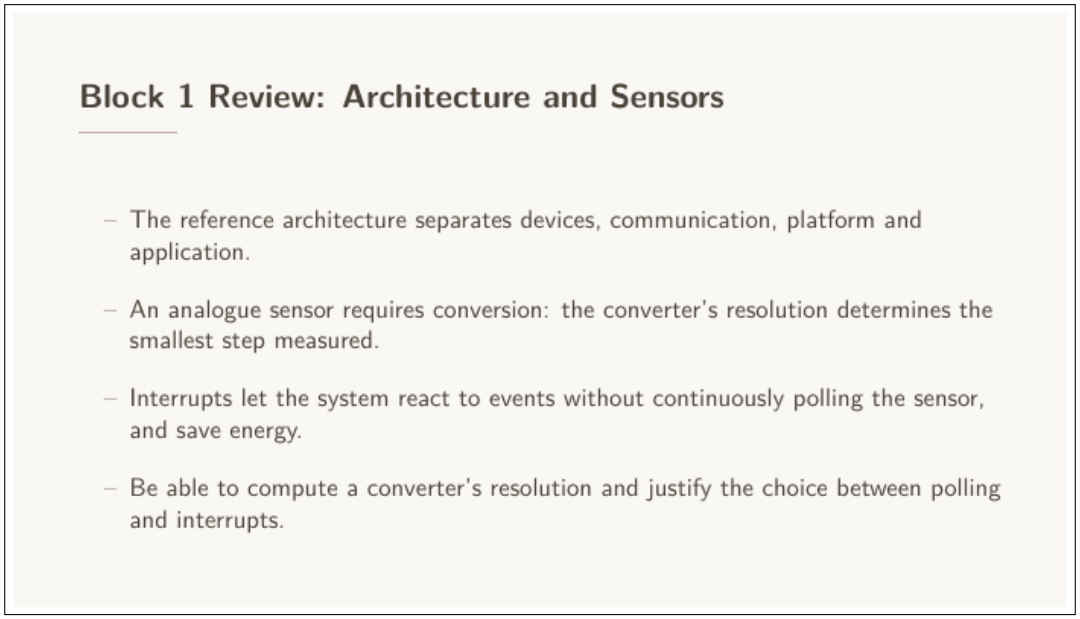
Network and application. The third block deals with IP networks for IoT, and the fourth with application and middleware. The order follows the stack, from the bottom up, and is the same the mini-project runs through in a single system.

Slide 5. Map of the Course



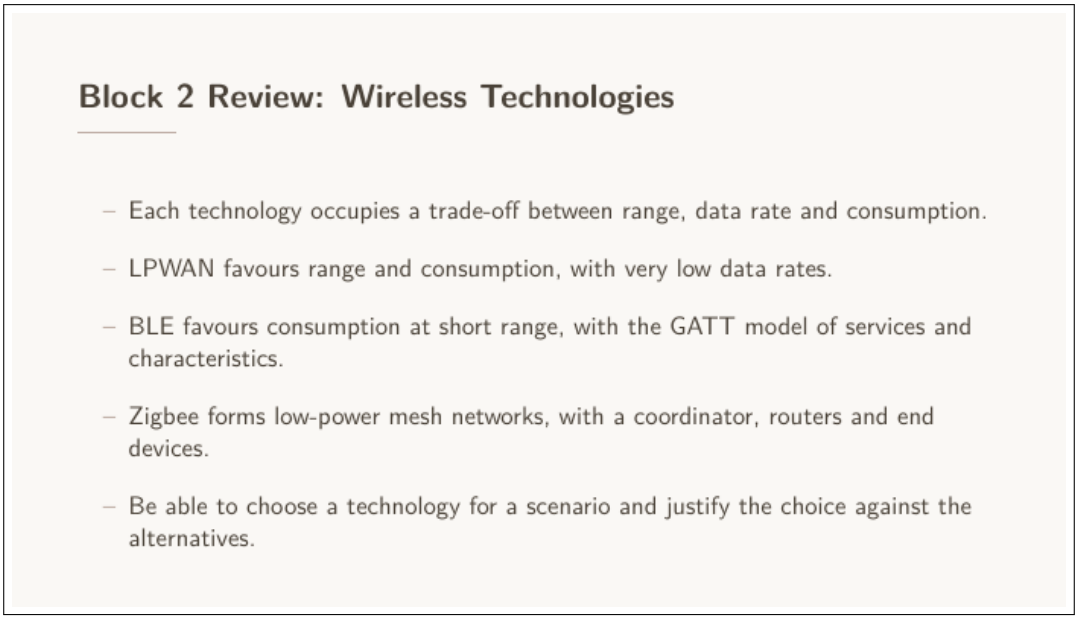
The course map. The figure stacks the four blocks in stack order, with the content of each. The student is encouraged to rebuild it from memory and associate with each block what was studied and what was done in the lab.

Slide 6. Block 1 Review: Architecture and Sensors



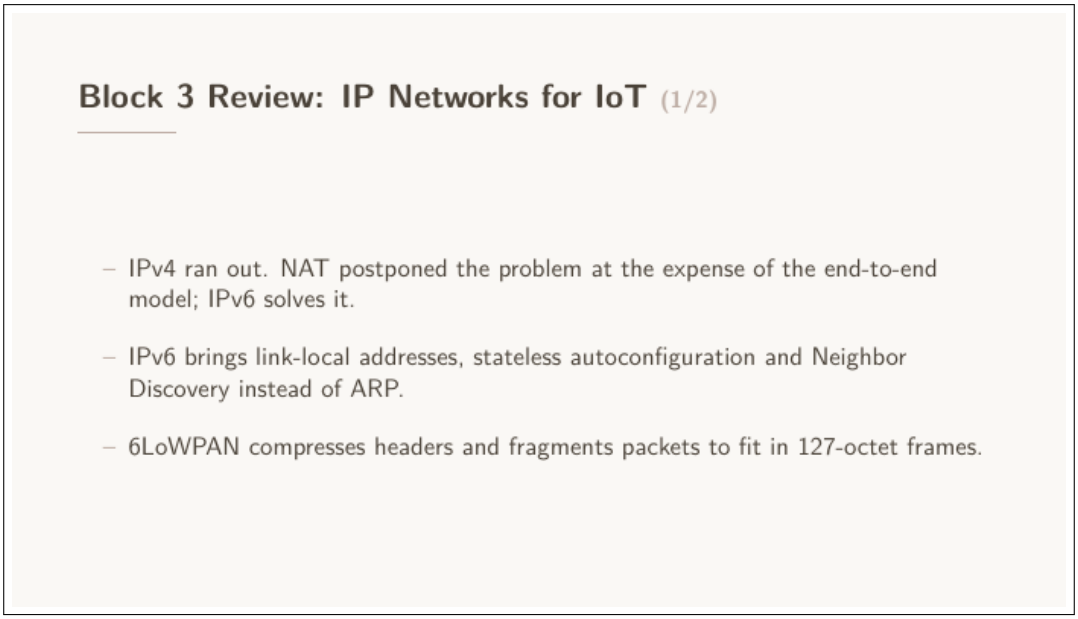
Architecture and sensors. A converter's resolution determines the smallest step measured, and the choice between polling and interrupts is a choice of energy and responsiveness. Check: what is the resolution of a 10-bit converter with a 5 V reference?

Slide 7. Block 2 Review: Wireless Technologies



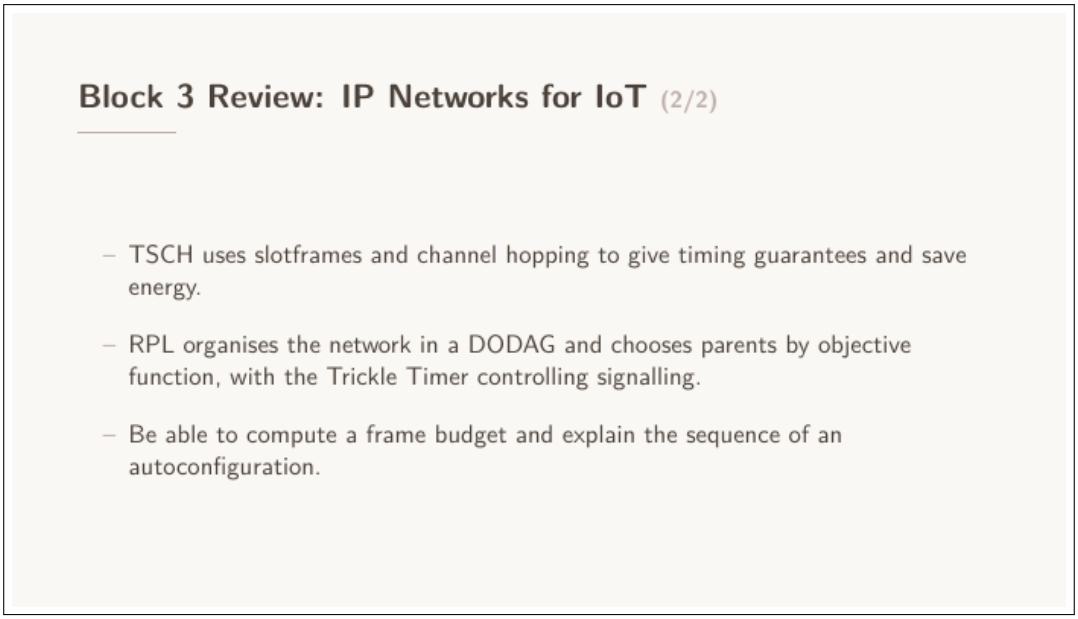
Wireless technologies. Each technology is a trade-off between range, data rate and consumption. The skill to demonstrate is choosing and justifying, not memorising values.

Slide 8. Block 3 Review: IP Networks for IoT (1/2)



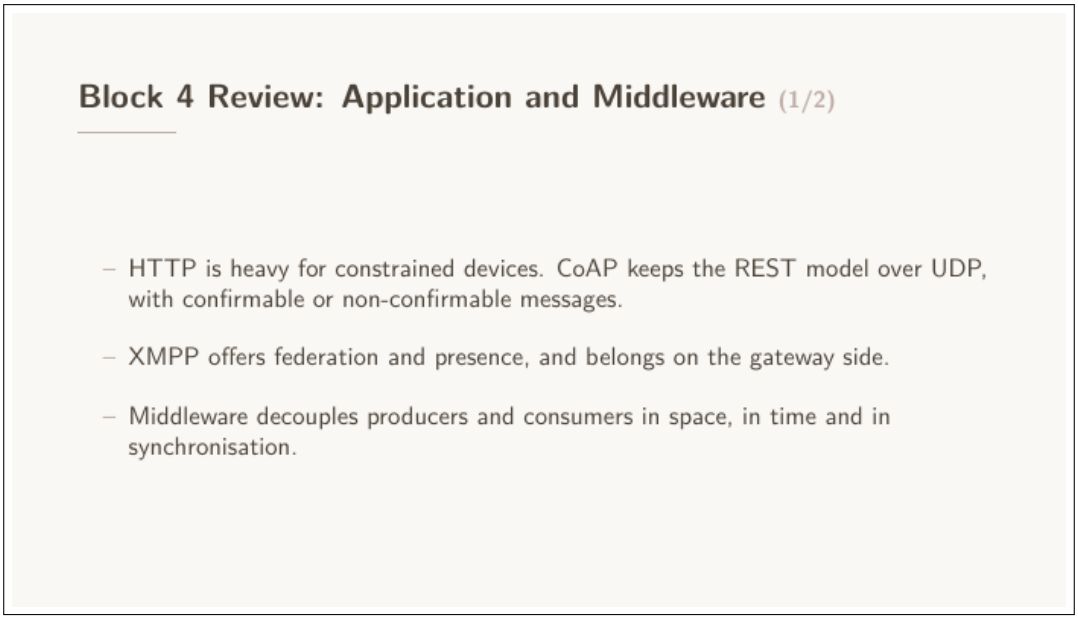
IPv6 and 6LoWPAN. IPv6 brings autoconfiguration and Neighbor Discovery. 6LoWPAN solves the problem of fitting in a small frame with compression and fragmentation. You should know the numbers: 127 octets, 48 down to 6, 54 and 33 of payload.

Slide 9. Block 3 Review: IP Networks for IoT (2/2)



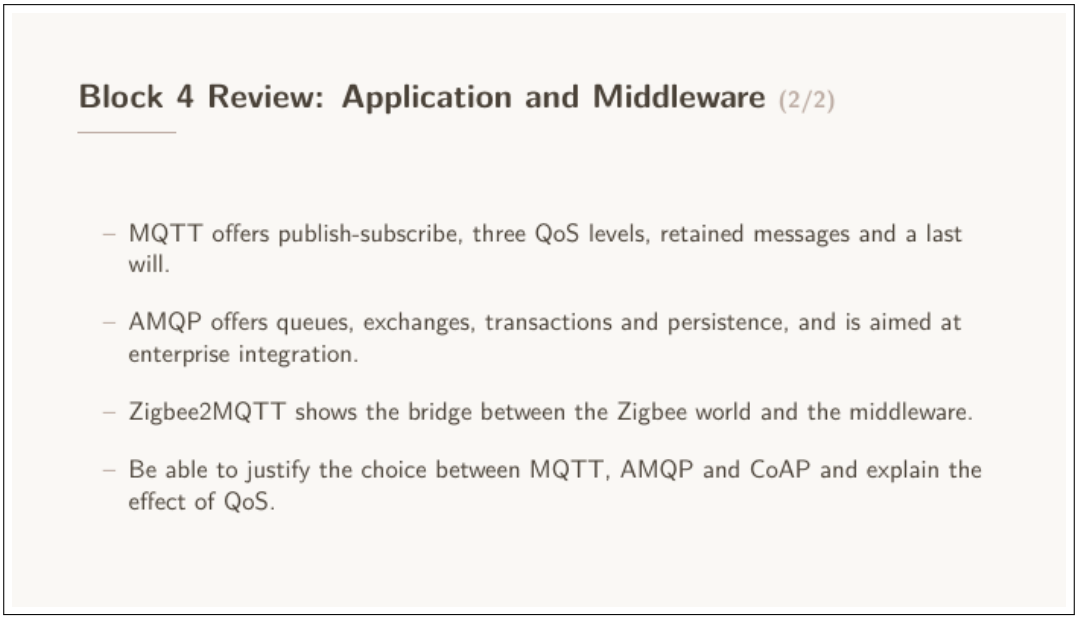
TSCH and RPL. TSCH gives a network time and channels, and RPL organises routes. You should be able to explain rank, objective function, ETX and the Trickle Timer.

Slide 10. Block 4 Review: Application and Middleware (1/2)



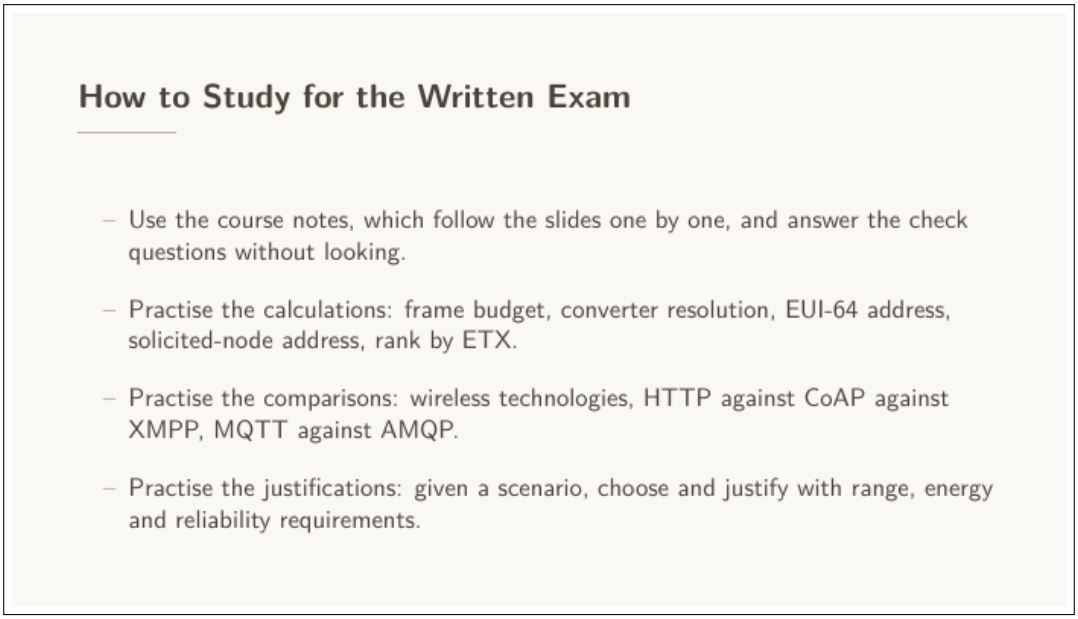
Application protocols. HTTP, CoAP and XMPP solve the same question in different ways, and the choice depends on transport, interaction model, encoding and weight.

Slide 11. Block 4 Review: Application and Middleware (2/2)



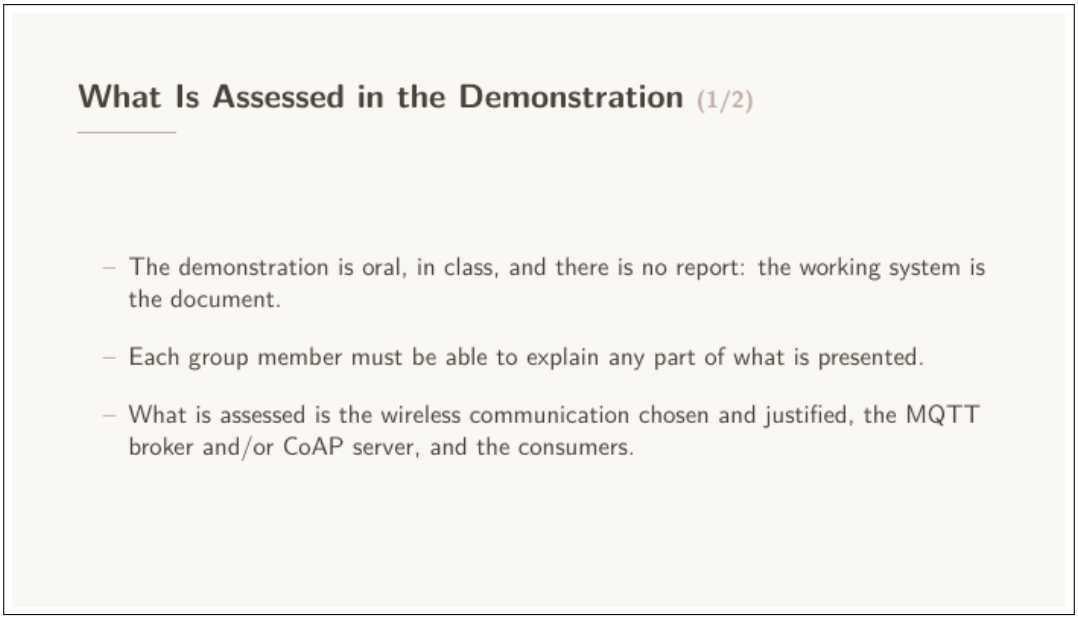
Middleware. MQTT and AMQP decouple producers and consumers. MQTT is minimal and constrained, AMQP is rich and enterprise-oriented. Zigbee2MQTT shows the link between a wireless technology and the middleware.

Slide 12. How to Study for the Written Exam



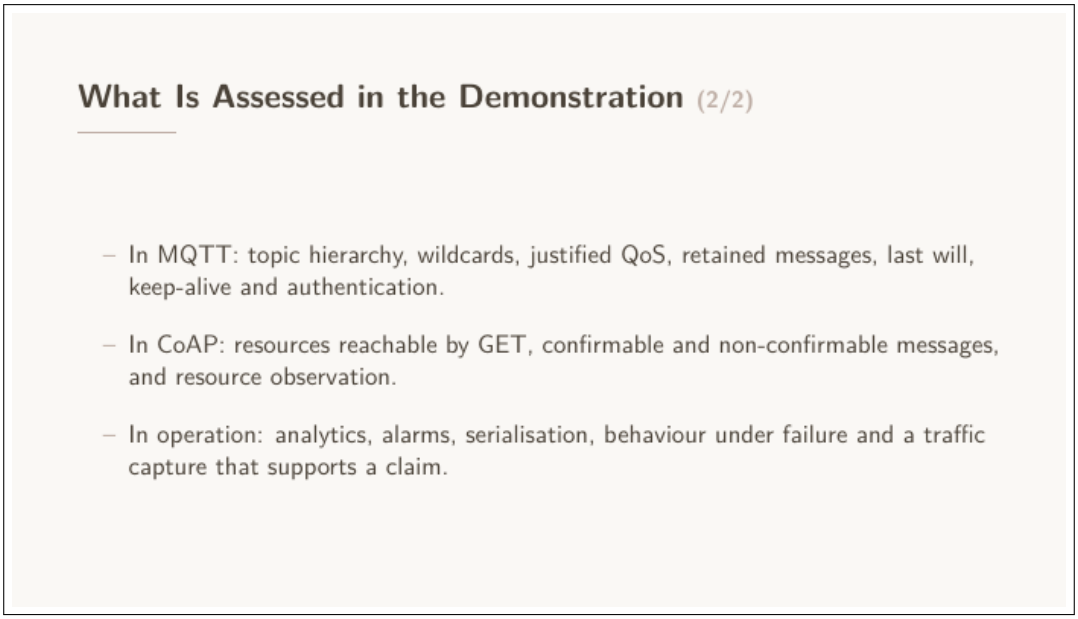
How to study. Four habits: use the course notes, practise the calculations, practise the comparisons and practise the justifications. A complete justification refers to the requirement, the technology and the trade-off accepted.

Slide 13. What Is Assessed in the Demonstration (1/2)



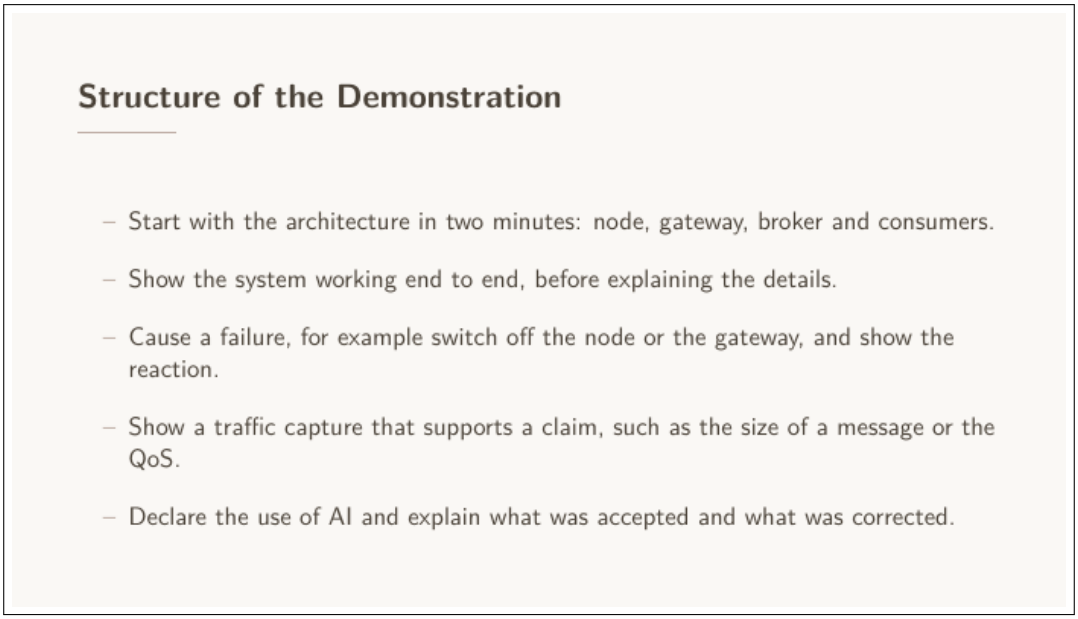
The demonstration is oral. There is no report. The working system is the document, and each group member has to be able to explain any part. The use of AI is permitted, but must be declared and is critically assessed.

Slide 14. What Is Assessed in the Demonstration (2/2)



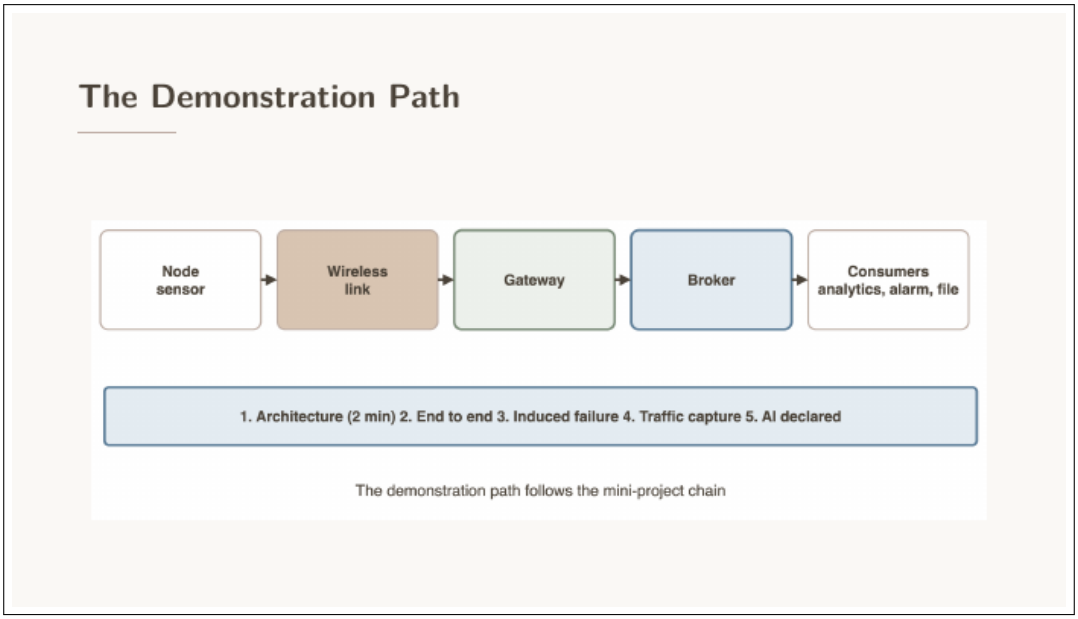
What is checked. The aspects to check are in the assignment and fall into wireless communication, MQTT broker, CoAP and operation. You should have a checklist and rehearse the demonstration against it.

Slide 15. Structure of the Demonstration



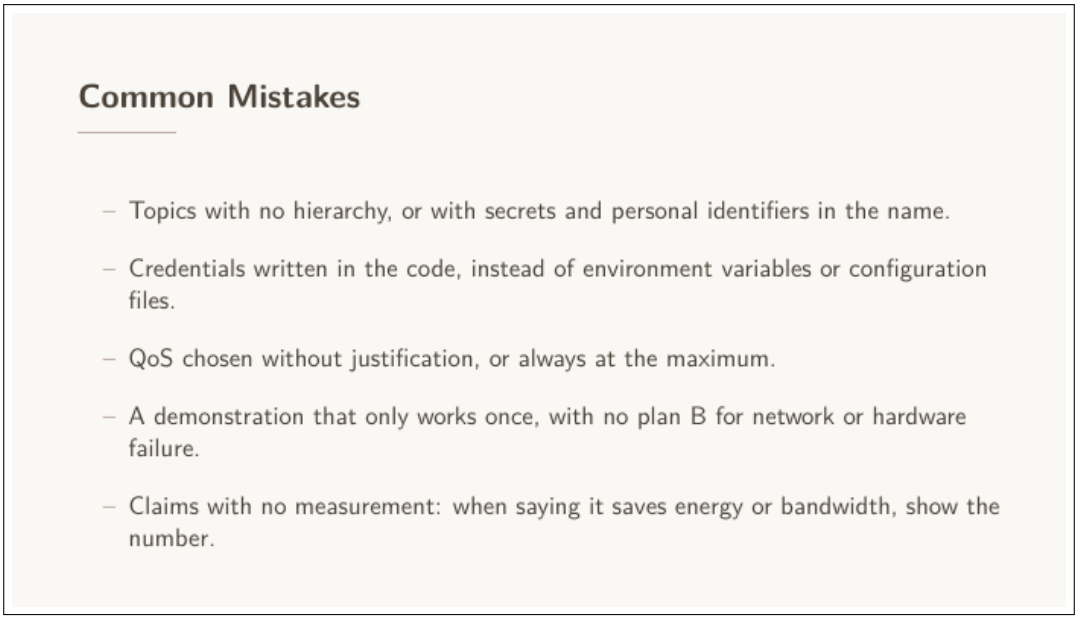
A demonstration in five steps. Architecture in two minutes, system working end to end, an induced failure, a traffic capture and the AI declaration. The order puts the system ahead of the explanation.

Slide 16. The Demonstration Path



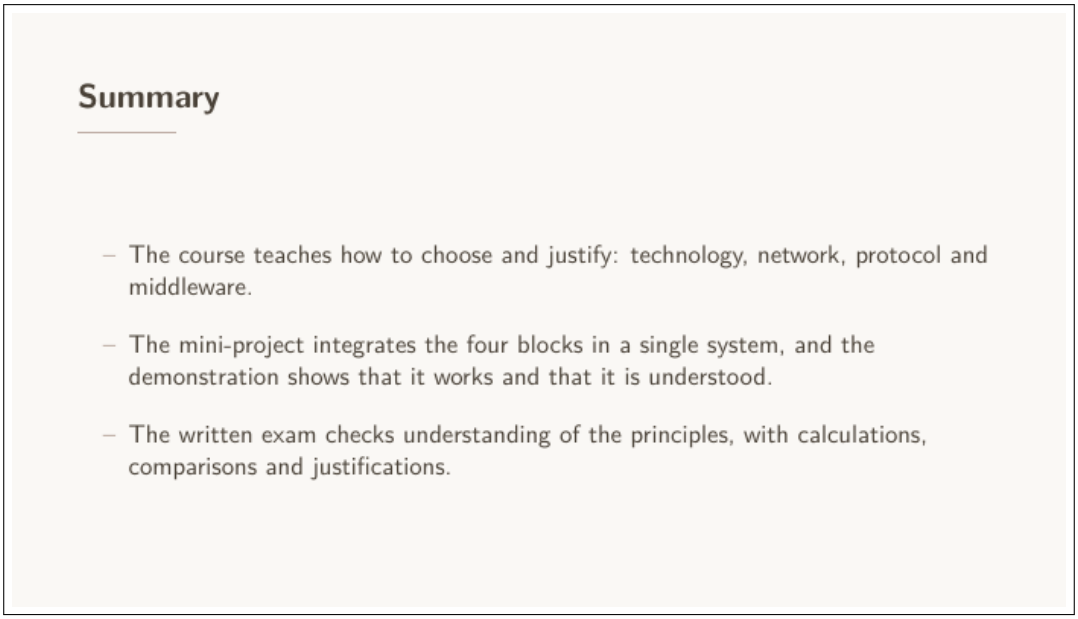
The demonstration path. The figure shows the chain from node to consumer and the five steps of the demonstration over it. Each step corresponds to an aspect the assessor looks for.

Slide 17. Common Mistakes



What usually goes wrong. Topics with no hierarchy, credentials in the code, QoS without justification, demonstrations that only work once and claims with no measurement. Each is easy to avoid with rehearsal.

Slide 18. Summary



The essentials. The course teaches how to choose and justify. The mini-project integrates everything, and the exam checks the principles. Whoever can explain why they chose what they chose has the essentials.

Final Check

Before considering the material studied, check that you can answer the following without looking at the slides.

1. Compute the number of payload octets in an 802.15.4 frame with a 25-octet MAC, IPv6 and UDP with no compression.
   Answer: $127 - 25 - 40 - 8 = 54$ octets.
2. Derive the EUI-64 identifier from the MAC 00:1A:2B:3C:4D:5E.
   Answer: 021A:2BFF:FE3C:4D5E (FFFE in the middle and the U/L bit flipped).
3. What is the solicited-node address of an address ending in 0E:8C6C?
   Answer: FF02::1:FF0E:8C6C.
4. Why does channel hopping improve reliability in TSCH?
   Answer: fading depends on frequency, and retransmitting on another frequency has a higher probability of success.
5. Which objective function chooses a more reliable parent, hops or ETX, and why?
   Answer: ETX, because it weighs link quality and not just their number.
6. Difference between CON and NON messages in CoAP.
   Answer: CON requires an ACK and retransmits with exponential back-off; NON does not wait for confirmation.
7. What distinguishes QoS 1 from QoS 2 in MQTT?
   Answer: QoS 1 guarantees at least one delivery and allows duplicates; QoS 2 guarantees exactly one, with four messages.
8. When would you choose AMQP over MQTT?
   Answer: when transactions, queues, persistence and integration between enterprise systems are needed.
9. What three decouplings does a broker offer?
   Answer: in space, in time and in synchronisation.
10. Which topics does `casa/+/temperatura` catch?
    Answer: the temperature of any room in casa, with exactly one level in the middle.
11. What are the pairing steps in Zigbee2MQTT?
    Answer: allow joining, put the device in pairing mode, interview, friendly name and close joining.
12. Why is AI use declared in the demonstration?
    Answer: because it is permitted but critically assessed, and everyone has to be able to explain what they present.

References

- Mini-project assignment. *TCIoT 2026-2027 Mini-Project.* Course material.
- Course notes and lab guides of Lectures 1 to 13.